Run report: join-q5-run-20261024T2238Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part JOIN under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: join-q5 20261024T2238Z
git HEAD: 7a579ce24983ef78f6af4ecf3dc9a8c6f10e8a91
tool: examples/genus4_n4_census.rs sha256 efce43f3fdaa848389a03f3d778a9b9604fe23d1d6e5e1bd75e777a49fb6cf01
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 65702f045ec07ff04edc651d7add4725ceecc7cf1bdc79b495a712a8cb942a44
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/join-q5-run-20261024T2238Z JOIN 5 research/genus4_n4_census_20261024/results/cen-q5-cone-run-20261024T2208Z/
mass_by_t.csv,research/genus4_n4_census_20261024/results/cen-q5-nonsplit-run-20261024T2210Z/mass_by_t.csv,research/genus4_n4_census_20261024/results/cen-q5-split-run-20261024T2206Z/mass_by_t.csv
research/genus4_n4_census_20261024/results/tian-q5-run-20261024T2237Z/tian_by_t.csv
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 0
```

tool_log.txt

```
q=5: prime-order traces with an elliptic curve: 14 (ordinary 13); traces with no hit: [49]
Spearman(mass, H) = 0.565
Spearman(mass, Haar density) = 0.775
Spearman(mass, Haar*H/sqrt(disc)) = 0.775
Spearman(mass, #l with Tian kernel (depends = 1/2)) = 0.288
fit mass = c*H: c = 1.317507e0, R^2 (through origin, ratio-of-sums c) = -0.028
fit mass = c*haar: c = 1.985322e2, R^2 (through origin, ratio-of-sums c) = 0.458
fit mass = c*haar*H/sqrt: c = 9.287875e2, R^2 (through origin, ratio-of-sums c) = 0.037
```

join_summary.txt

```
q=5: prime-order traces with an elliptic curve: 14 (ordinary 13); traces with no hit: [49]
Spearman(mass, H) = 0.565
Spearman(mass, Haar density) = 0.775
Spearman(mass, Haar*H/sqrt(disc)) = 0.775
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fit mass = c*H: c = 1.317507e0, R^2 (through origin, ratio-of-sums c) = -0.028
fit mass = c*haar: c = 1.985322e2, R^2 (through origin, ratio-of-sums c) = 0.458
fit mass = c*haar*H/sqrt: c = 9.287875e2, R^2 (through origin, ratio-of-sums c) = 0.037
```

join_by_t.csv (first 12 of 14 rows)

t	nE	ordinary	hits	mass	mass_se	H	haar_density	arith_x_haar	tian_yes	tian_depends	tian_no
−47	673	true	28	0.7417	0.1402	4.000	0.014330	0.003360	2	0	1
−33	659	true	1037	15.9690	0.5486	4.000	0.064274	0.006844	3	0	0
−27	653	true	2036	36.6702	0.8797	8.000	0.077702	0.014771	3	0	0
−21	647	true	781	13.6464	0.5328	8.000	0.086747	0.015294	2	0	1
−17	643	true	901	13.7084	0.5054	16.000	0.090489	0.030791	2	0	1
7	619	true	403	10.2439	0.5107	8.000	0.080361	0.012986	1	0	2
9	617	true	523	12.4208	0.5595	8.000	0.077446	0.012597	3	0	0
13	613	true	799	14.2213	0.5474	20.000	0.070895	0.029368	2	1	0
19	607	true	323	6.6719	0.3956	8.000	0.059547	0.010300	2	0	1
25	601	false	1070	22.9753	0.7374	12.333	0.046875	0.013351	2	1	0
27	599	true	637	9.9517	0.4385	8.000	0.042467	0.008073	3	0	0
33	593	true	81	1.9807	0.2201	4.000	0.029089	0.003098	3	0	0

Graph

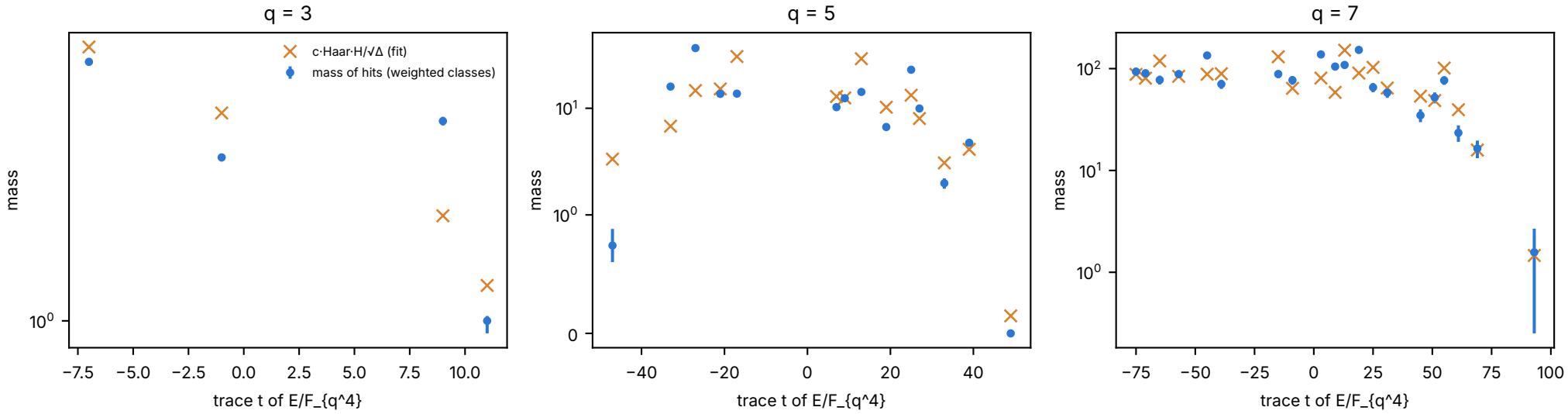


Figure 1: Mass of hits per prime-order trace with the Haar-density fit.